

ATS RACE Workflow

ATS RACE release Checklist(c to D sample drawing release)

Release Points		Result/Status	Deadline	Resp.
Design	1. Final 3D model and offer drawing officially confirmed by customer?	done	2025.5.14	卢高松
	2. Final nameplate confirmed by customer?	done	2025.5.14	卢高松
	3. IMDS or CAMDS finished?	done.		
	4. D-FMEA closed and signed ? Sign?	done	2025.5.14	卢高松
	5. I-FMEA signed with customer? Sign?	done		
	6. Installation instruction finished?	done		
MOF	7. Final SC/CC finished by EGT5/MOE/QMM/PUR? Sign?	done	2025.5.14	卢高松
	8. Catalyst spec, offer drawing, PSC, TCD all available in BOM?	done	2025.5.14	卢高松
	1. Final welding process flow chart finished by MOE?			
Validation	2. Welding and assembly trial run finished?	ongoing	2025.6.24	卢高松
	3. P-FMEA signed ?			
	1. 9000 Km bad road Vehicle validation status?	已做51%预计6月10完成		
System	2. PV report?	Covered by DV	2025.5.14	卢高松
	1. Customer validation report? (If customer take responsible the calibration, take as reference)	done		卢高松
	2. TCD is available?	done	2025.5.14	卢高松
Flow	3. PD is available?	done		
	1. QG0 and QG1 passed? QGC2 plan?	done	11	卢高松

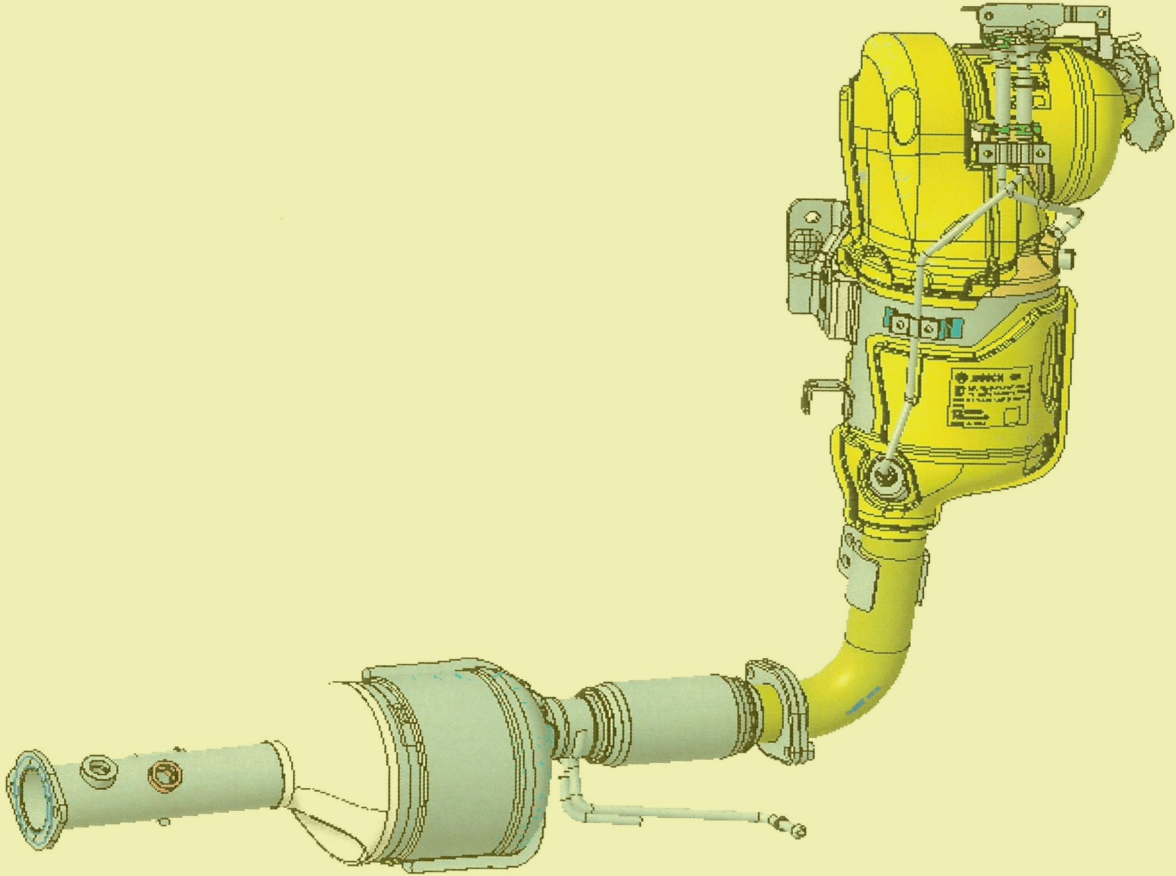
**: Finished w/o open points; RACE can be proceeded.

Finished w/ open points and countermeasures are available. Escalation to SM and inform to PM for RAC release or not.

Finished w/ open points but root cause is not clear and countermeasure is not available; Inform PM and reject the RACE with inputting the reason in reject step.

***: Design SPIJM responsible for the checklist document, but relative points should be checked with the Resp. person.



BOSCH 博世 Confidential	Product Development Design Change Request 产品开发设计变更请求		Version 15 Effective date:2025-05-13th	
	Design Change Request No. 25_045			
Customer project Name/ 客户项目名称: JIM-493 MCR No. / MCR号:				
Sample status / 样件状态: ☐ A Sample ☐ B Sample ☐ C Sample ☒ D Sample				
Change part&product Name/更改零部件产品的名称(简要描述)				
DOC-DPF: F01ZH003FU-00 SCR: F01ZH003FV-00 BRACKET: F01Z5102EU-02				
Reason of changes/更改理由 (简要描述): C样件放行				
Change from/变更来源: Customer request / 客户要求 ☒ Supplier request / 供应商请求 ☒ Design optimization / 设计优化 ☒ Correction / 更正(错误) ☐ Other / 其他 ☐				
Change inform to/变更通知人: Zhang Shengchang,Zhang Hanquan,Shen Weibo, Li Yicen, Ge Jilong Yang Guangtuo, Luo jing, Lu Qingsong				
Whether this change impact others component or product 此变更是否影响到其他部件和产品: ☐ yes/是 ☒ no/否				
If impact others product, Whether others projects&product change or not? 如果影响, 别的部件和产品是否也需要同时变更?				
Design / 设计工程师 Signature / 签名: 李青松 Date / 日期: 2025.5.14 Check / 校对工程师 闫松 Signature / 签名: Date / 日期: 2025.5.14				
Description of Change/更改描述: (下面空白处请列出详细的变更前后对比) 详见变更描述PPT 				

Confidential

Design Change
Request No.

25_045

1. Function&Performance will be influenced? / 系统性能影响?

☐ yes/是☒ no/否

If Yes, System impact evaluation?

如果是, 系统影响评估?

(如果是初次样件放行, 空白处请说明样件目的)

AISPOL 已确认

Whether need sample to test? Plan?

是否需要样件进行验证? 计划?

NA

System/签名

沈伟波

Whether the catalyst information is confirmed?

催化剂信息是否确认? (物料号、图纸、SPEC)

Catalyst/签名

2. Interface&boundary (internal and customer) will be influenced? / 接口&边界(内部和客户)影响?

☐ yes/是☒ no/否

If impact customer, whether get confirm from customer side?

请说明影响和是否得到客户的确认?

Design/签名

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3. Mechanical strength&Durability will be influenced? / 机械强度&可靠性影响?

☐ yes/是☐ no/否

If Yes, How to evaluate? Plan?

如果是, 请说明影响和如何评估, 计划?

(例如: 仿真分析等)

结构强度同原设计

Whether need sample to test? Plan?

是否需要样件进行验证? 计划?

Validation/签名

卢青松

4. Product document influenced?

4.1 Interface FMEA-relevant / IFMEA:

☐ no/否☒ yes/是

Note

4.2. Product FMEA-relevant / DFMEA:

☐ no/否☒ yes/是

Note

4.3. Special Characteristics /PSC: (针对C Sample)

☐ no/否☒ yes/是

Note

4.4. IMDS relevant: (针对C Sample)

☐ no/否☒ yes/是

Note

4.5 Offer drawing relevant: (针对C Sample)

☐ no/否☒ yes/是

Note

4.6. TCD relevant:

☐ no/否☒ yes/是

Note

Design/签名

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5. Changed parts Stock& treatment / 需要变更零件的库存统计和处理?

(只针对涉及到已经生产过的零件变更, 新放行零件和本条不相关)

☐ NA. (新零件放行, 不涉及库存)☒ 涉及到已经生产过的零件变更

RBCW RDC(外库)和RBCW 仓库, 是否有库存,

☐ 无库存☐ 有库存, 库存数量

供应商处(已做好, 未发货)是否有库存,

☐ 无库存☐ 有库存, 库存数量

LPN工厂是否有库存,

☐ 无库存☐ 有库存, 库存数量

在途, 运输中是否有库存

☐ 无库存☐ 有库存, 库存数量

PUE/签名

被影响到的实物库存处理指示:
(设计工程师组织会议讨论)☐ 可以继续使用完☐ 改制后使用☐ 报废

Design/签名

报废成本 改制费用: ☐

PUE/签名

已装车件处理方法(发客户或标定车等)
(SMP组织会议讨论)其他(单独说明): ☐

改制

更换: ☐

6. Influence on supplier part?

Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier?
变更后的零件, 生产可行性供应商是否确认? 反馈?

设计设计讨论

Influence on purchasing part delivery time? 对采购件交货时间的影响?

涉及到变更的零件, 博世已出PR, 供应商还未来得及生产的订单, 是否已经通知供应商取消.

☐ no/否 ☒ yes/是

Does the change influence on purchasing cost? 变更对采购成本的影响?

☐ increase/增加, 金额☒ decrease/降低, 金额☐ no change/不变

PUE/签名

卢青松

7. DFMA & Sample production OPL?

必要时附加DFMA 文件

Design/签名

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8. Influence on LPN manufacturing&assembly process?

Feedback from manufacturing engineer? 生产制造装配是否影响, 反馈?

Influence on project&product delivery time? 对项目&产品交货时间的影响?

D样量产未820

Influence on manufacturing cost / 对生产成本的影响?

☐ increase/增加, 金额☐ decrease/降低, 金额☐ no change/不变

PUE/签名

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9. PM alignment? (Influence on project Timeline? Cost?);

是否同意对项目时间和成本的影响: 需求样品订单数量

PM/签名

卢青松

10. 如果涉及以下情况需TCR3 审核确认. 不涉及则 NA:

☐ 库存报废, 金额

平均成本增加, 金额

生产成本增加, 金额

RBCW/TCR3

Design Team Leader

ENG-CV1

PUR

TCR3

2025-06-20

Signature/签名

卢青松

Signature/签名

卢青松

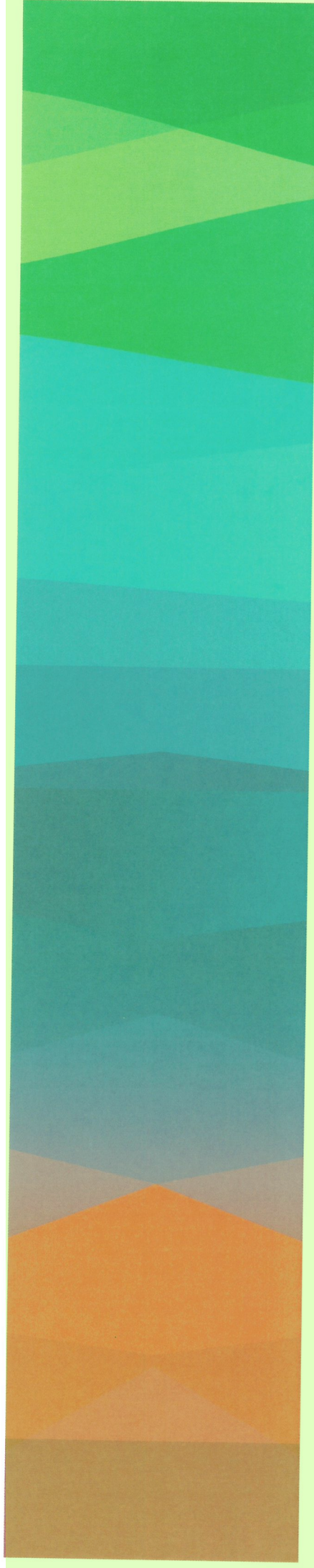
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卢青松

Signature/签名

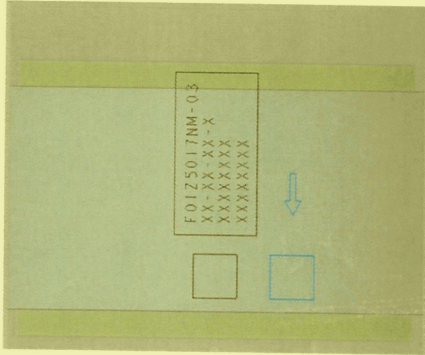
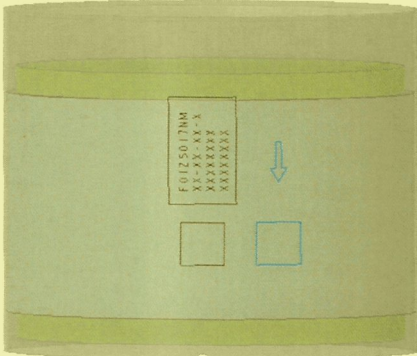
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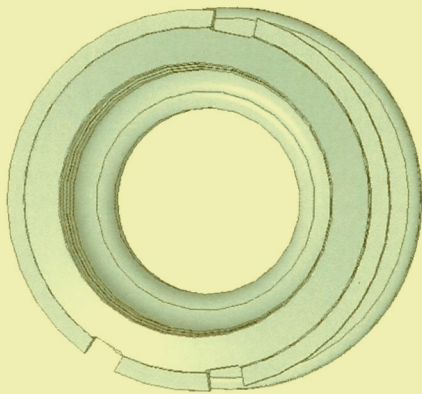
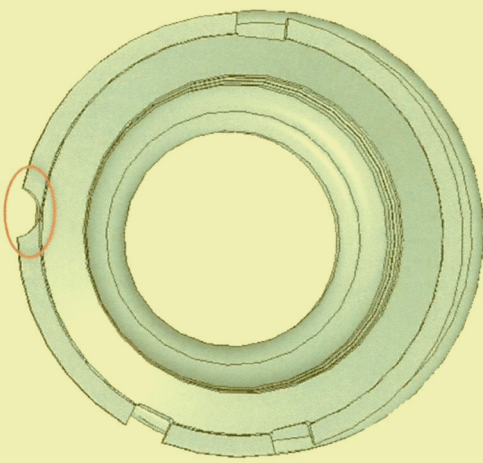
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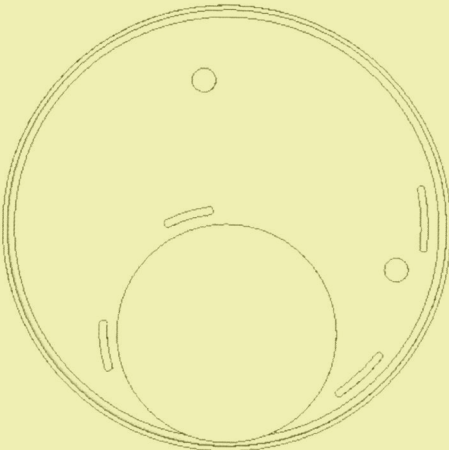
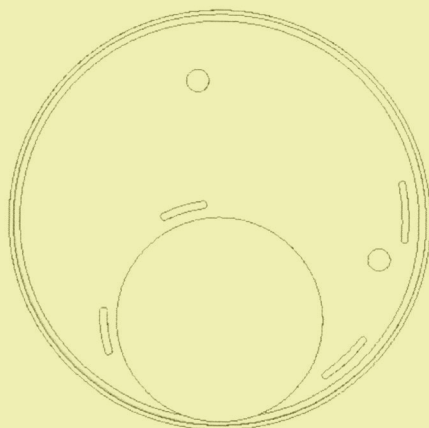


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ENG-CV

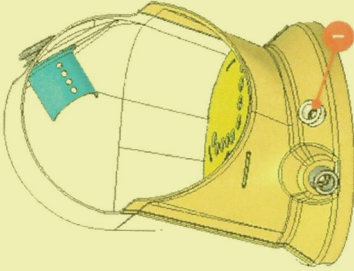
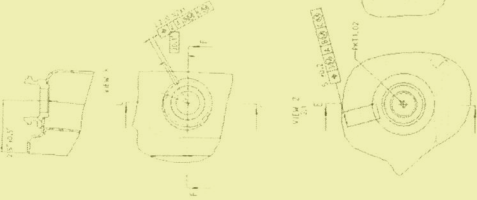
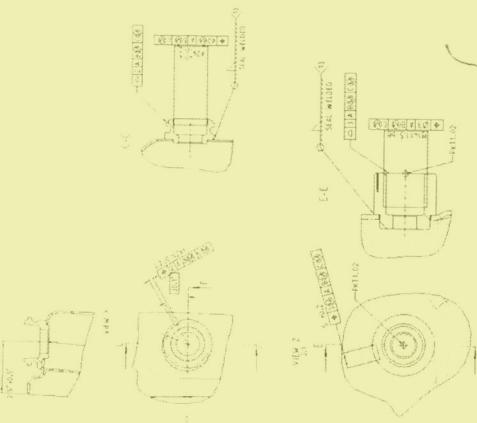
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	变更后					
						
	DOC CANNING: F01Z5017NM-03 DOC CANNING SLEEVE: F01Z5102ED-02		DOC CANNING: F01Z5017NM-04 DOC CANNING SLEEVE: F01Z5102ED-03			
	DOC筒体刻字离焊缝太近, 加大15mm据焊缝的距离;					
	变更原因	无				
	风险点 (影响)	无				
验证方法	无					

变更描述	零件号		零件名称	Project	样件阶段	变更来源
	F01Z5102E0-01		DM flange	JIM-493	D	BOSCH/Customer
	变更前					变更后
						
	F01Z5102E0-00			F01Z5102E0-01		
变更原因	供应商反馈JIM 493与4JJ尿素喷嘴座只有槽的角度不同，因此加上防错缺口；					
风险点（影响）	无					
验证方法	无					

变更描述	零件号	零件名称	Project	样件阶段	变更来源
	F01Z51027K	PLATE	JIM-493	D	BOSCH/Customer
	变更前				
					
	F01Z5101BP		F01Z51027K		
	变更原因				
	供应商请求混合器内隔板换成JP560量产产内隔板;				
风险点（影响）					
无					
验证方法					
无					

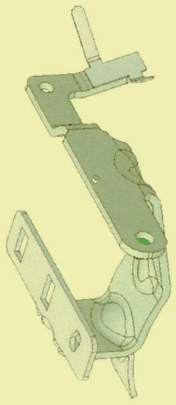
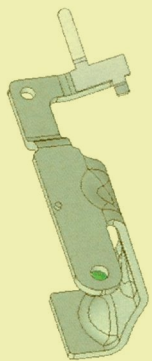
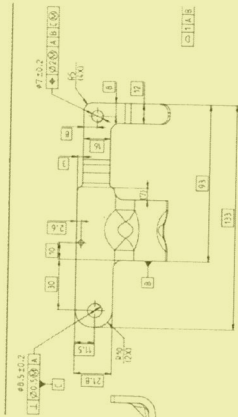
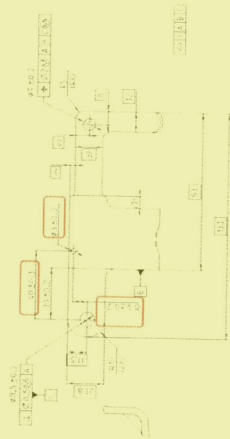
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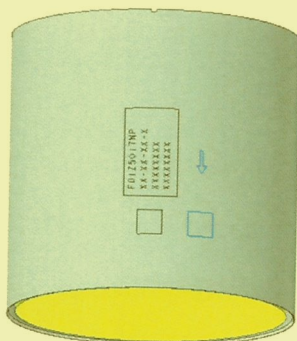
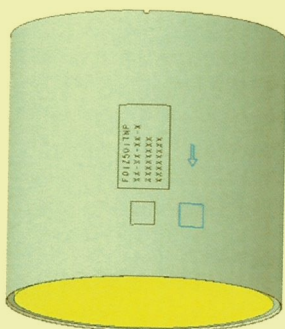
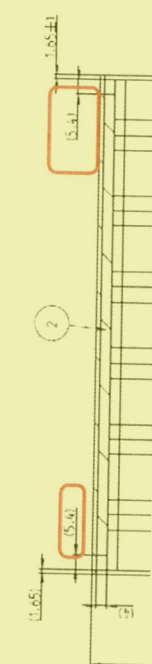
变更描述	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017W6-02	MIXER UNIT	JIM-493	D	BOSCH/Customer
	变更前				
变更描述					
	MIXER UNIT: F01Z5017W6-01 PLATE: F01Z5101BP DM flange:F01Z5102E0-00 DP SENSOR BOSS:F01Z510265-00				
变更原因	1.把压差座转到焊接总成上; 2.混合器组件图纸部分尺寸转移到焊接组件上;		MIXER UNIT: F01Z5017W6-02 PLATE: F01Z51027K DM flange:F01Z5102E0-01		
风险点 (影响)	1.根据供应商实际生产状态, 把压差座转到焊接总成上; 2.根据供应商检具, 把混合器组件图纸上的部分尺寸转移到焊接组件上;		无		
验证方法	无		无		

1

3

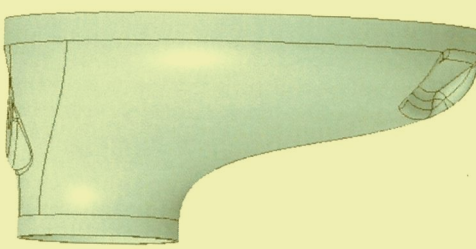
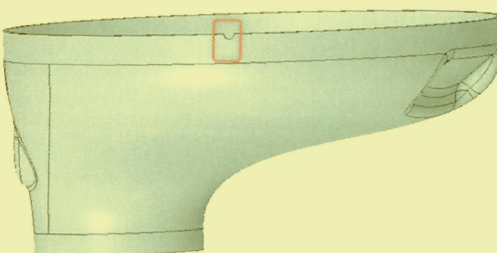
零件号		零件名称		Project	样件阶段	变更来源
F01Z5017X1-04		DP TUBE BRACKET		JIM-493	D	BOSCH/Customer
变更前						
变更描述						
						
	DP TUBE BRACKET WITH NUT: F01Z5017X1-04 DP TUBE BRACKET: F01Z51026E-02 SENSOR BRACKET : F01Z5102DH-04			DP TUBE BRACKET WITH NUT: F01Z5017X1-05 DP TUBE BRACKET: F01Z51026E-02		
	变更原因			1.客户反馈线束固定支架超温，取消线束固定支架；2.更新压差支架零件和组件的图纸，把压差支架的凸点直径和位置标注上；		
	风险点（影响）			无		
验证方法			无			

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	零件号	零件名称	Project	样件阶段	变更来源																																																																		
	F01Z5017NP-03	DPF CANNING	JIM-493	D	BOSCH/Customer																																																																		
	变更前		变更后																																																																				
变更描述	<table data-bbox="509 1108 761 1449"><tr><td>Substrate type</td><td>inch</td><td>50PF</td></tr><tr><td>Substrate size</td><td>inch</td><td>6.77x7.7</td></tr><tr><td>Catalyst weight</td><td>kg</td><td>2.86</td></tr><tr><td>Back pressure</td><td>kPa</td><td>18.5</td></tr><tr><td>Acceleration load</td><td>g</td><td>48.770±0.6</td></tr><tr><td>Substrate material</td><td></td><td>SiC</td></tr><tr><td>Substrate strength</td><td>MPa</td><td>1.7</td></tr><tr><td>Part gap</td><td>mm</td><td>4</td></tr><tr><td>Load tolerance</td><td>mm</td><td>±0.5</td></tr><tr><td>Canning process</td><td></td><td>Non-GBU</td></tr><tr><td>Sleeve material</td><td></td><td>643</td></tr></table> A-A  DPF CANNING: F01Z5017NP-02 DPF MAT: F01Z5102EA-01 <table data-bbox="509 188 802 647"><tr><td>Substrate type</td><td>inch</td><td>50PF</td></tr><tr><td>Substrate size</td><td>inch</td><td>6.77x7.7</td></tr><tr><td>Catalyst weight</td><td>kg</td><td>2.86</td></tr><tr><td>Back pressure</td><td>kPa</td><td>18.5</td></tr><tr><td>Acceleration load</td><td>g</td><td>48.770±0.6</td></tr><tr><td>Substrate material</td><td></td><td>SiC</td></tr><tr><td>Substrate strength</td><td>MPa</td><td>1.7</td></tr><tr><td>Part gap</td><td>mm</td><td>4</td></tr><tr><td>Load tolerance</td><td>mm</td><td>±0.5</td></tr><tr><td>Canning process</td><td></td><td>Non-GBU</td></tr><tr><td>Sleeve material</td><td></td><td>643</td></tr></table> A-A  DPF CANNING: F01Z5017NP-03 DPF MAT: F01Z5102EA-02					Substrate type	inch	50PF	Substrate size	inch	6.77x7.7	Catalyst weight	kg	2.86	Back pressure	kPa	18.5	Acceleration load	g	48.770±0.6	Substrate material		SiC	Substrate strength	MPa	1.7	Part gap	mm	4	Load tolerance	mm	±0.5	Canning process		Non-GBU	Sleeve material		643	Substrate type	inch	50PF	Substrate size	inch	6.77x7.7	Catalyst weight	kg	2.86	Back pressure	kPa	18.5	Acceleration load	g	48.770±0.6	Substrate material		SiC	Substrate strength	MPa	1.7	Part gap	mm	4	Load tolerance	mm	±0.5	Canning process		Non-GBU	Sleeve material		643
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变更原因	1.采购确认量产衬垫为奇耐，DPF衬垫在原基础上减短5mm，避免露棉；2.衬垫改为居中放置；																																																																						
风险点（影响）	无																																																																						
验证方法	无																																																																						

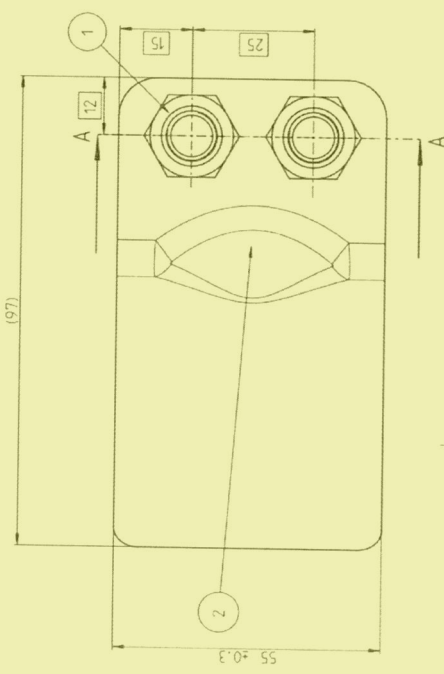
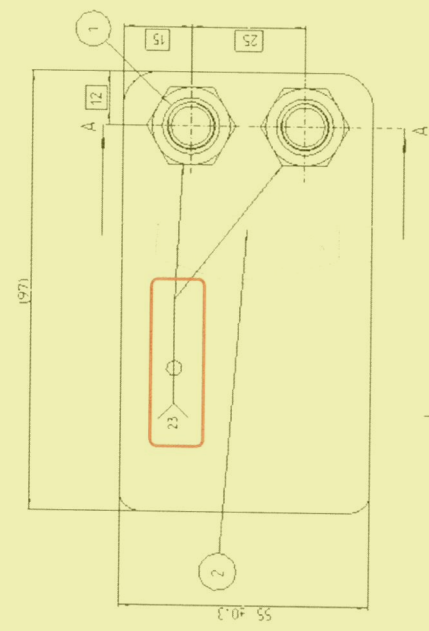
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3

变更描述	零件号		零件名称	Project	样件阶段	变更来源
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	变更前					变更后
	F01Z51025W-02					
	F01Z51025W-03					
	F01Z51025W-02		F01Z51025W-03			
	F01Z51025W-03		F01Z51025W-02			
变更原因	供应商请求在DPF出气端锥上增加定位缺口；					
风险点（影响）	无					
验证方法	无					

PDECR 25-045

3

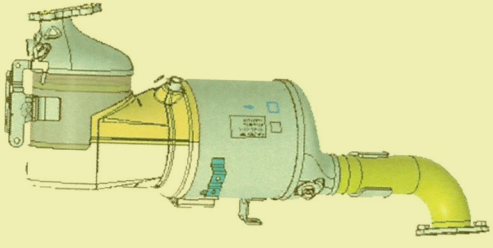
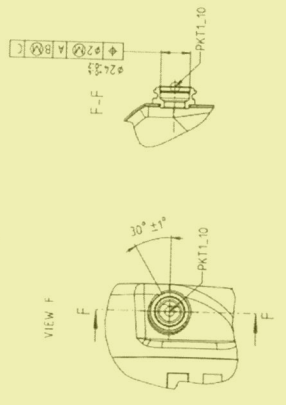
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	变更前					
						
	F01Z5017XR-02			F01Z5017XR-02		
	变更原因					
	DPF出气尾管支架图纸上增加电阻焊;					
风险点 (影响)						
无						
验证方法						
无						

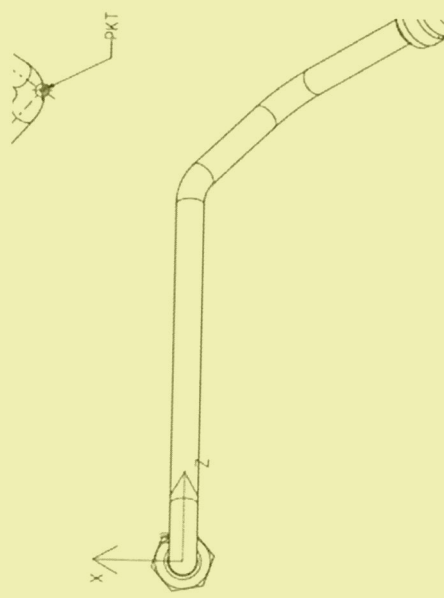
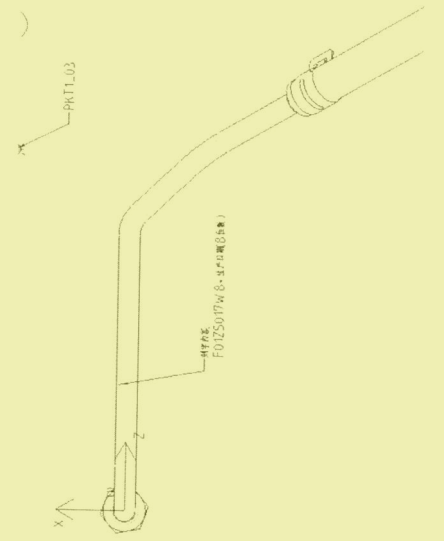
1

2



BOSCH

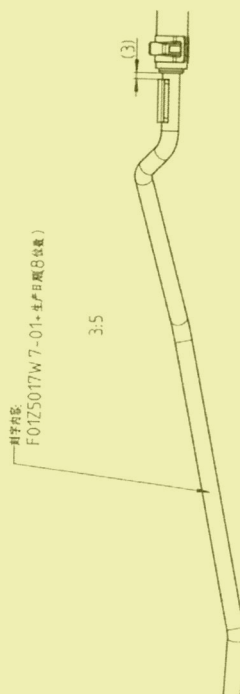
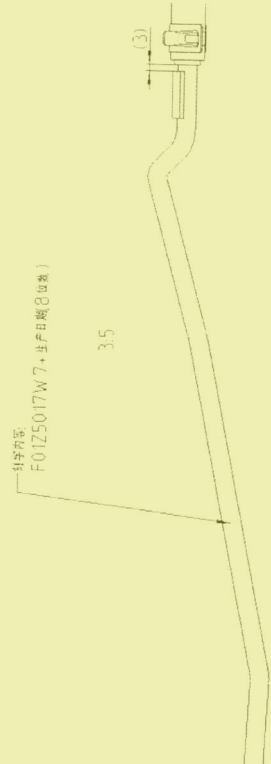
	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017W5-05	DOC+SDPF WELDING	JIM-493	D	BOSCH/Customer
	变更后				
变更描述	 F01Z5017W5-04  F01Z5017W5-05				
变更原因	更新焊接组件图纸，尿素喷嘴角度改为位置度方便检具检测；				
风险点（影响）	无				
验证方法	无				

零件号		零件名称	Project	样件阶段	变更来源
F01Z5017W8-02		DP tube inlet	JIM-493	D	BOSCH/Customer
		变更前	变更后		
变更描述					
		F01Z5017W8-02	F01Z5017W8-02		
变更原因		新增进气压产管刻字;			
风险点 (影响)		无			
验证方法		无			

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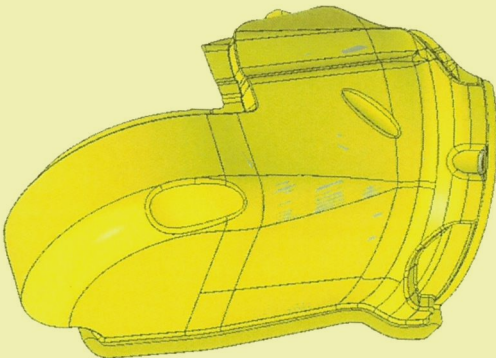
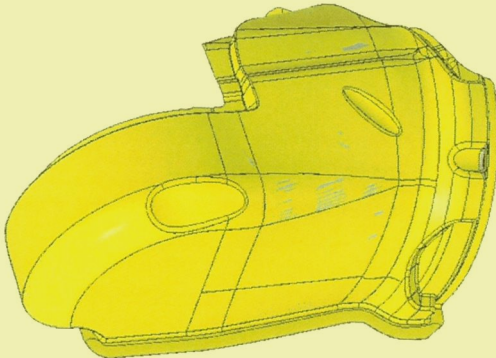
2

零件号		零件名称	Project	样件阶段	变更来源
F01Z5017W7-02		DP tube inlet	JIM-493	D	BOSCH/Customer
		变更后			
变更描述		 			
		F01Z5017W7-02			
变更原因		更改出气压产管刻字;			
风险点 (影响)		无			
验证方法		无			

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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017NU-02	FOIL	JIM-493	D	BOSCH/Customer
变更描述	变更前		变更后		
					
变更原因	供应商反馈保温壳开裂，报废率高，请求把430材质更改为304；				
风险点（影响）	无				
验证方法	无				

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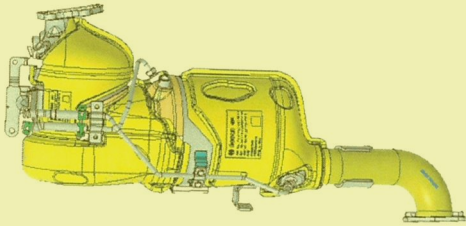
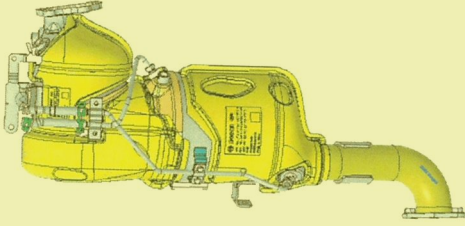
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零件号		零件名称	Project	样件阶段	变更来源
See Below		MAT	JIM-493	D	BOSCH/Customer
变更前		变更后			
变更描述		F01Z5102RG-02 F01Z5102RJ-02 F01Z5102RK-01 F01Z5102RL-01 F01Z5102RM-01 F01Z5102RN-01 F01Z5102RP-00 F01Z5102RR-01 F01Z5102RT-01 F01Z5102RU-00 F01Z5102RZ-01		F01Z5102RG-02 F01Z5102RJ-02 F01Z5102RK-01 F01Z5102RL-01 F01Z5102RM-01 F01Z5102RN-01 F01Z5102RP-00 F01Z5102RR-01 F01Z5102RT-01 F01Z5102RU-00 F01Z5102RZ-01	
变更原因		保温棉图纸按照最终适配更新图纸;			
风险点 (影响)		无			
验证方法		无			

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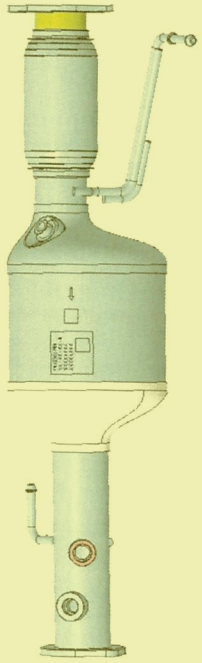
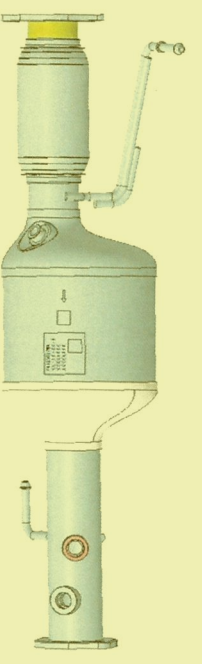
	零件号	零件名称	Project	样件阶段	变更来源
	F01ZH003FU-00	DOC+SDPF UNIT	JIM-493	D	BOSCH/Customer
变更描述	变更前				
	变更后				
	F01ZH003FU-06				
变更原因	上述更新后, DOC+SDPF总成随之更改;				
风险点 (影响)	无				
验证方法	无				

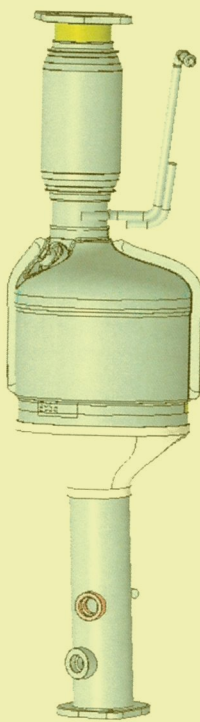
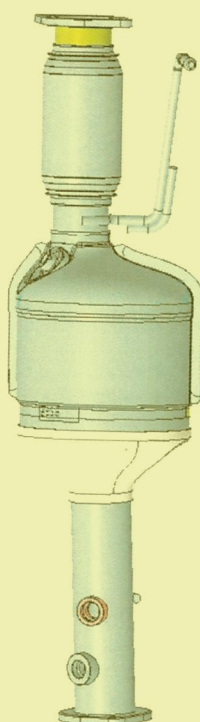


变更描述	零件号		零件名称		Project		样件阶段		变更来源		
	F01Z5017WA-03		SCR CANNING		JIM-493		D		BOSCH/Customer		
变更原因	变更前		变更后								
	F01Z5017WA XX-XX-XX-X XXXXXXXXXX XXXXXXXXXX Substrate type Substrate size Catalyst weight back pressure Acceleration Load Substrate material Substrate Strength Mat gap Gap tolerance Canning Process Sleeve material SCR 6.77x5 1.59 5 20 Condienite 1 3.5 ±0.5 Non-G30 4.41		F01Z5017WA XX-XX-XX-X XXXXXXXXXX XXXXXXXXXX Substrate type Substrate size Catalyst weight back pressure Acceleration Load Substrate material Substrate Strength Mat gap Gap tolerance Canning Process Sleeve material SCR 6.77x5 1.59 5 20 Condienite 1 3.5 ±0.5 Non-G30 4.41 SCR CANNING: F01Z5017WA-02 SCR MAT: F01Z5102EB-00 SCR CANNING: F01Z5017WA-03 SCR MAT: F01Z5102EB-01								
	采购确认量产衬垫为奇耐，SCR衬垫图纸更改为原图纸，衬垫更改后SCR封装随之更改；										
	无										
	无										

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2

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017W9-04	SCR WELDING	JIM-493	D	BOSCH/Customer
变更描述	变更前		变更后		
					
变更原因	F01Z5017W9-03		F01Z5017W9-04		
风险点 (影响)	上述更改后, SCR焊接总成随之更改;				
验证方法	无		无		

变更描述	零件号	零件名称	Project	样件阶段	变更来源
	F01ZH003FV-00	SCR UNIT	JIM-493	D	BOSCH/Customer
	变更前				
变更描述					
	F01ZH003FV-06		F01ZH003FV-00		
变更原因	上述更改后，SCR总成随之更改；				
风险点（影响）	无				
验证方法	无				

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零件号		零件名称		Project	样件阶段	变更来源
F01Z5018C0-03		INSULATION		JIM-493	D	BOSCH/Customer
		变更前		变更后		
变更描述						
	F01Z5018C0-02		F01Z5018C0-03			
	变更原因		样件图号更新后, 保温刻字信息随之更改;			
	风险点 (影响)		无			
验证方法		无				

3

2